

DexSwarm AI

Native Swarm Intelligence System

Rethinking data, models and robots to build machine teams



Robot-Free Multimodal Capture · Swarm Data Collection

Collaboration, built in from the start

About Us

Wuhan Lingwo Intelligent Technology Co., Ltd. (DexSwarm AI) explores and builds next-generation native swarm intelligence.

Starting with real-world team tasks, we co-develop swarm data, intelligence models and collaborative robots, enabling robots to understand roles, divide work and complete tasks together.

Our robot-free multimodal capture devices and data platform support real-world collaboration data capture and team-task modeling.

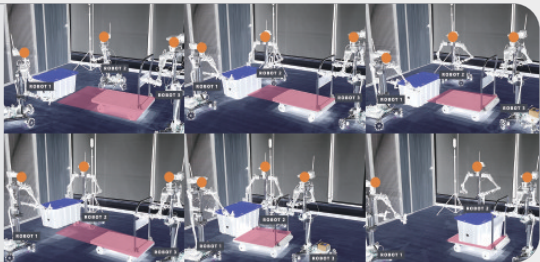


Three Layers THREE SYSTEM LAYERS

01 Swarm Data

SWARM DATA

Capture multi-agent actions and coordination: waiting, responding and handovers.



Current
Products

02 Swarm Models

SWARM INTELLIGENCE

Understand roles and states; study task allocation, decisions and action toward shared goals.



Future
R&D Roadmap

03 Team Robotics

COLLABORATIVE ONTOLOGY

Design robots for reliable communication, flexible networks and coordinated execution.

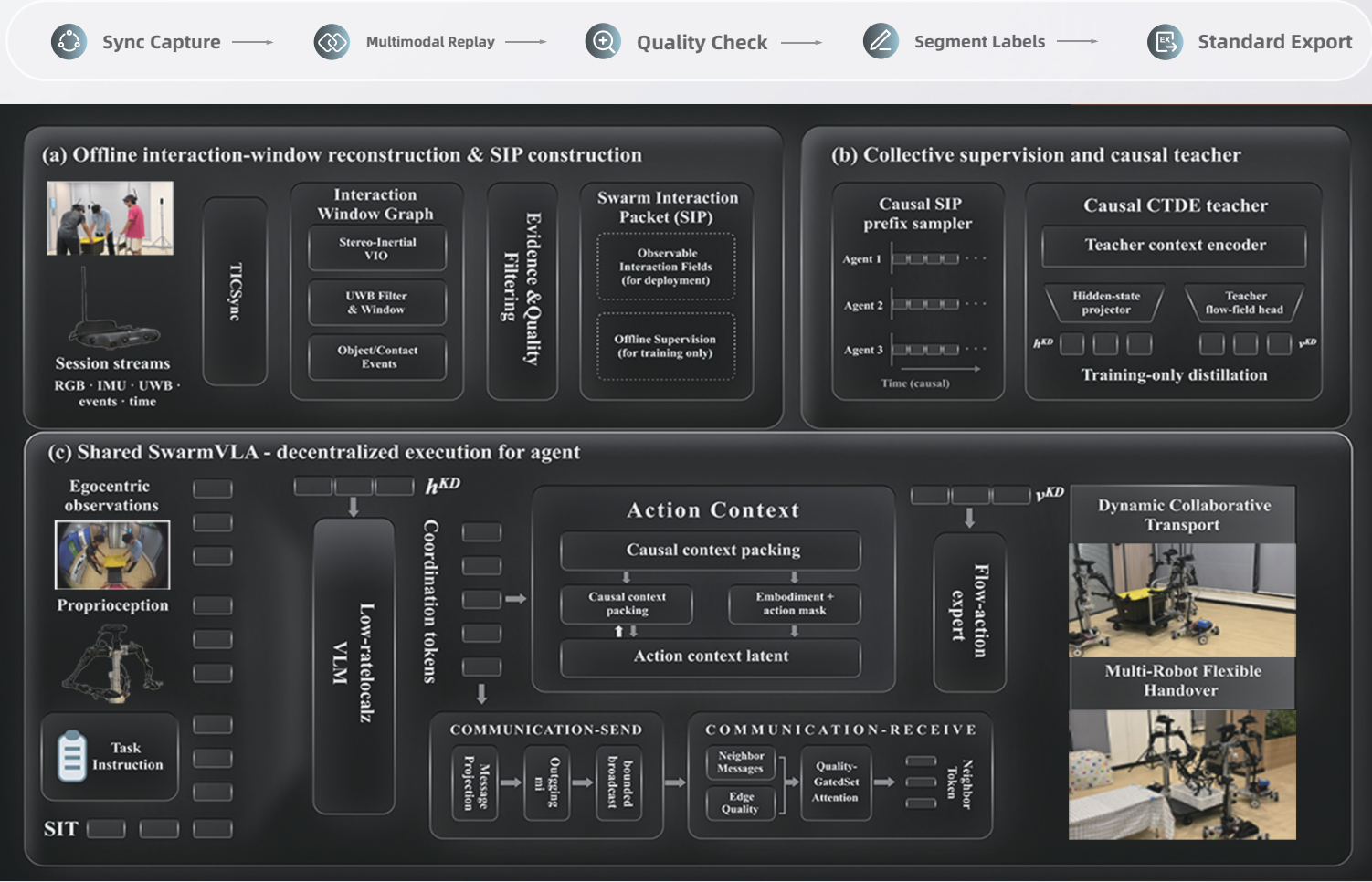


“Native means **collaboration**
in data, models and robots from day one.”

DexSwarm Model (DSM)

Decentralized Swarm Intelligence Model

Software Workflow



Closed-Loop Pipeline





Turn Real Collaboration into Swarm Data

Record collaboration. Reuse experience. Create value from data.

Make collaboration recordable, make experience reusable, and let data create value

01 Collaboration Data

Capture real collaboration across devices and views.
Structure actions, trajectories and environment data.



Multimodal Capture



Synchronized Teams



Reliable Transfer

Motion / vision / force / environment Concurrent multi-person capture Motion / vision / force / environment
Synchronized multimodal capture Real collaboration and interaction Synchronized multimodal capture

02 Multi-Robot Operation

Robots jointly perform complex tasks with
real-time sync and precise, reliable coordination.



Group Ego Data



Robot Adaptability



Joint Operation

Multi-person egocentric capture Swarm-driven policy optimization Joint robot control and collaboration
Reconstruct shared task perspectives Improve robot adaptability Extend operations across spaces

Capture Multimodal Experience from Real Human Actions.

Real-world sensing. Precision capture. Multimodal fusion. Efficient replay.



DexSwarm Ego



First-person capture of surroundings and head position

Visual Input	6 cameras: task, scene and localization vision
FOV per Camera	206°FOV
Head Tracking Accuracy	Mean error ≤ 8 mm; max. ≤ 16 mm (vs. optical motion capture)
Visual Resolution	1920x1080 30Hz
Multimodal Sync Error	< 4 ms
Battery Life	> 3 h; hot-swappable batteries



UMI Dex 1



Capture dexterous hand kinematics, wrist pose, vision and touch

Hand Kinematics	Ultra-precise EMF; 21 DOF per hand
Capture Modalities	6D wrist pose; wrist vision; touch; hand kinematics
Hand Tracking Accuracy	Mean error < 8 mm; max. ≤ 16 mm (vs. optical motion capture)
Visual Input	1920x1080 30Hz; 206°FOV
Multimodal Sync Error	< 4 ms
Battery Life	> 2 h; hot-swappable batteries



UMI G1



Capture two-finger gripper paths, jaw opening and contact data

Overview	Matched capture and execution gripper design
Capture Modalities	6D pose; wide-angle vision; touch; jaw opening
Pose Tracking Accuracy	Mean error ≤ 8 mm; max. ≤ 16 mm (vs. optical motion capture)
Visual Input	1920x1080, 30Hz; 206°FOV
Multimodal Sync Error	< 4 ms
Battery Life	> 2 h; hot-swappable batteries



Companion Gripper



Matched hardware links capture to robot execution

Actuation	Servo motor drive
Max. Grip Force	45 N
Max. Stroke	90 mm
Visual Input	1920x1080, 30Hz; 206°FOV
Repeatability	± 0.03 mm
Communication	Modbus RTU + expandable I/O



Real-World Sensing
Recreate real operations



Precision Capture
Submillimeter positioning



Multimodal Fusion
Vision, touch and motion



Efficient Replay
Low latency, high fidelity

From Robots That Perform Tasks to Robots That Work as a Team

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